



NWU®

NORTH-WEST UNIVERSITY
NOORDWES-UNIVERSITEIT
YUNIBESITI YA BOKONE-BOPHIRIMA

CMPG325-COMPUTER NETWORKS

Milestone 1: CLIENT DESIGN REVIEW

Student	Muthambi, R
Student Number	46657665
Project ID	CMPG325-2026-099
Client ID	CLI-099
Assigned Organization	Mafikeng Local Municipality Satellite Office
Industry	Municipal Services
Repository	GITHUB

Date: 28 AUGUST 2026

1. Introduction

1.1 Project Background

This document constitutes the Milestone 1 Client Design Review submission for the CMPG 325 Computer Networks individual semester project (Project ID CMPG325-2026-099). The project requires the analysis of an assigned client scenario, the design and simulation of a suitable network in Cisco Packet Tracer, the configuration of an assigned networking challenge, and the documentation of evidence throughout the project lifecycle.

The project follows a structured workflow beginning with client-problem analysis and proceeding through requirements analysis, network design, IP address planning, implementation in Packet Tracer, configuration of the assigned feature, testing, documentation and video demonstration. Milestone 1 corresponds to the design phase of this workflow and precedes full implementation and verification.

1.2 Client Overview

The assigned client is the Mafikeng Local Municipality Satellite Office (Mahikeng), operating within the Municipal Services industry (Client ID CLI-099). The satellite office provides several public-facing and internal administrative functions that must be supported by a segmented and controlled data network.

1.3 Purpose of the Network Design

The purpose of this design is to propose network architecture that connects the satellite office's operational areas while enforcing the client's stated security constraint, accommodating an approved client change request, and satisfying the assigned static-routing technical challenge. The design is developed around the assigned IPv4 addressing block 192.168.43.0/24, as required by the project brief.

2. Client Requirements

2.1 Business Requirements

The satellite office comprises five functional areas that must be supported by the network: Public Counter, Licensing & Permits, Municipal Administration, Finance, and a Shared Print Services facility. Each area has distinct access requirements, with the administrative and finance functions requiring a greater degree of protection than the public-facing counter and licensing functions.

2.2 Network Requirements

The following requirements are drawn directly from the official project brief and must be satisfied by the design:

- Use Cisco Packet Tracer for the network implementation and simulation.
- Design an appropriate topology and device arrangement for the assigned scenario.
- Use the assigned addressing block, 192.168.43.0/24, as the basis of the IP addressing plan.
- Configure the necessary routers, switches, end devices and other required nodes.
- Demonstrate successful connectivity and testing.
- Document significant design decisions and evidence in GitHub.

2.3 Security Constraint

The project brief specifies a single, non-negotiable design constraint: “**Public counter staff must not access the administrative network.**” This constraint is treated as the primary driver of the network's logical segmentation. It is addressed by placing the Public Counter and Municipal Administration functions in separate VLANs and subnets, with the intention of enforcing the restriction through an access control list once implementation begins. Section 6 discusses this in further detail.

2.4 Assigned Technical Challenge

The assigned networking challenge is Static Routing (multi-router path control), classified as intermediate difficulty. The design must configure, verify and demonstrate static routes that allow the router-connected networks to communicate across multiple hops.

2.5 Client Change Request

The client has raised Change Request CR8, requiring a shared printer zone to serve two departments that currently cannot print. The proposed design provides a centralized Shared Print Services network (VLAN 50) for Licensing & Permits (VLAN 20) and Municipal Administration (VLAN 30). PRINT-01 is connected through MUN-R2, allowing both departments to access the shared resources across the routed network

3. Proposed Network Design

3.1 Design Overview

The proposed design uses a three-router backbone in which each router serves one physical area of the satellite office. This arrangement was selected specifically so that the assigned static-routing challenge could be meaningfully demonstrated: traffic between the outer routers (MUN-R1 and MUN-R3) must be forwarded across the middle router (MUN-R2), creating genuine remote networks rather than a single flat broadcast domain.

3.2 Physical Topology

The physical backbone consists of three routers connected in series by point-to-point serial links, each router connected to a local access switch:

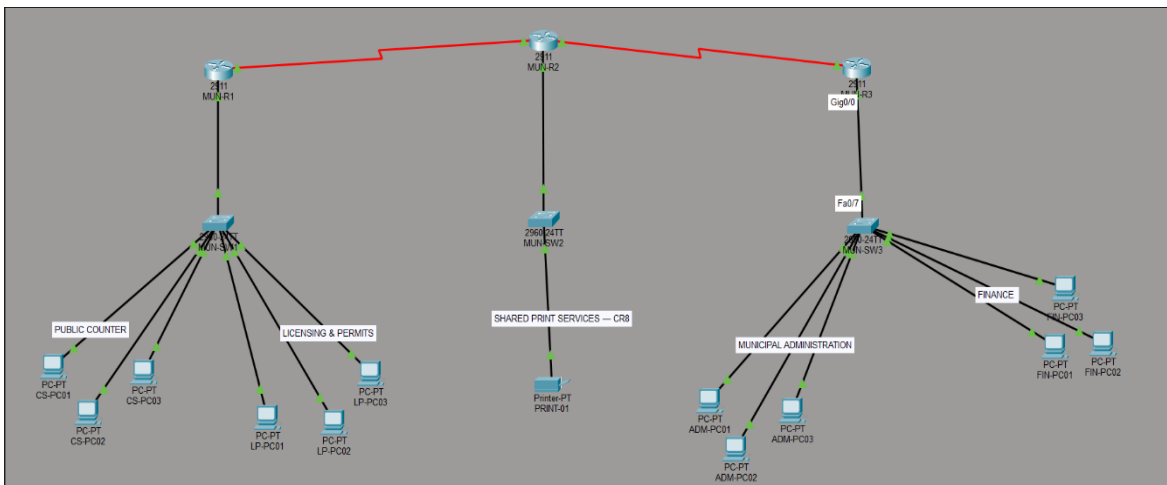


Figure 1: Proposed physical topology of the satellite office network.

- MUN-R1 connects to MUN-SW1, which serves the Public Counter devices (CS-PC01 to CS-PC03) and the Licensing & Permits devices (LP-PC01 to LP-PC03).
- MUN-R2 connects to MUN-SW2, which provides the centralized Shared Print Services network containing PRINT-01.
- MUN-R3 connects to MUN-SW3, which serves the Municipal Administration devices (ADM-PC01 to ADM-PC03) and the Finance devices (FIN-PC01 to FIN-PC03).
- The point-to-point serial links (MUN-R1 ↔ MUN-R2 ↔ MUN-R3) mean that any communication between a device on MUN-R1's networks and a device on MUN-R3's networks must traverse MUN-R2, providing an appropriate environment for demonstrating static, multi-router path control.

3.3 Device Inventory

The current topology comprises 19 devices and endpoints, summarized below:

Category	Quantity	Device names
Routers	3	MUN-R1, MUN-R2, MUN-R3
Switches	3	MUN-SW1, MUN-SW2, MUN-SW3
Public Counter PCs	3	CS-PC01, CS-PC02, CS-PC03
Licensing & Permits PCs	3	LP-PC01, LP-PC02, LP-PC03
Administration PCs	3	ADM-PC01, ADM-PC02, ADM-PC03
Finance PCs	3	FIN-PC01, FIN-PC02, FIN-PC03
Shared printer	1	PRINT-01
Total	19	—

Table 1: Complete device inventory for the proposed network.

3.4 Logical Topology

The logical topology overlays VLANs, subnets, default gateways and security boundaries onto the physical backbone described above. Five logical networks are proposed, corresponding to the four municipal departments and the shared print service:

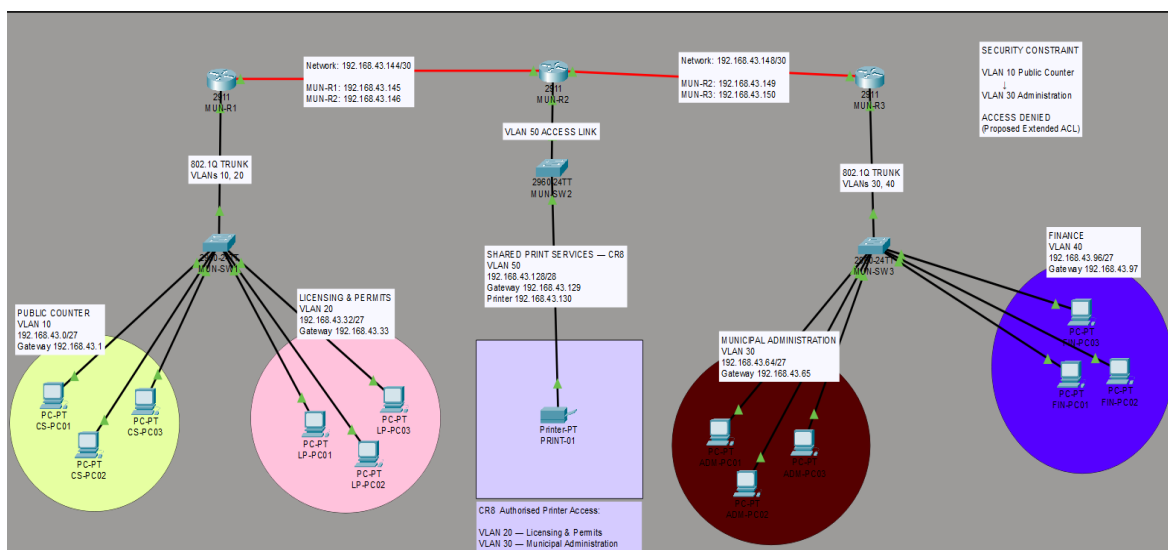


Figure 2: Logical relationship between department networks and the shared printer zone.

3.5 VLAN Design

Each department or service zone is assigned its own VLAN, providing a distinct broadcast domain and a clear point at which access control can later be applied. Subnetting the assigned 192.168.43.0/24 block in this way provides logical departmental separation, reduces the size of each broadcast domain, simplifies traffic management, directly supports the client's security constraint, keeps address allocation organized, and leaves capacity for future devices within each department.

VLAN	Department / Zone	Proposed subnet	Gateway
10	Public Counter / Citizen Services	192.168.43.0/27	192.168.43.1
20	Licensing & Permits	192.168.43.32/27	192.168.43.33
30	Municipal Administration	192.168.43.64/27	192.168.43.65
40	Finance	192.168.43.96/27	192.168.43.97
50	Shared Print Services	192.168.43.128/28	192.168.43.129

Table 2: Proposed VLAN and subnet allocation.

3.6 Router Interconnection

The network uses three routers connected in a linear topology: MUN-R1 → MUN-R2 → MUN-R3. Two /30 point-to-point networks are allocated for these links. The MUN-R1–MUN-R2 connection uses 192.168.43.144/30, while the MUN-R2–MUN-R3 connection uses 192.168.43.148/30. This design creates the multi-router path required for the assigned Static Routing challenge while efficiently using the available IPv4 address space.

4. IP Addressing Plan

4.1 Assigned Address Block

The client has been assigned the IPv4 address block 192.168.43.0/24. All internal subnets in this design are derived from this block using Variable Length Subnet Masking (VLSM), in line with the project brief's requirement that the assigned block form the basis of the addressing plan.

4.2 Subnetting

Departmental networks are sized as /27 subnets, each providing 30 usable host addresses, which comfortably accommodates the three PCs currently allocated to each department together with room for growth. The Shared Print Services zone is sized as a smaller /28 subnet, providing 14 usable addresses, which is sufficient for the printer and any future shared peripherals without allocating an entire /27 to a single-device zone. The two router-to-router links are sized as /30 subnets, the most efficient allocation for a point-to-point connection that requires only two usable host addresses.

4.3 Subnet Summary Table

VLAN	Department	Network	Subnet mask	Usable host range	Gateway	Broadcast
10	Public Counter / Citizen Services	192.168.43.0/27	255.255.255.224	192.168.43.1 - 192.168.43.30	192.168.43.1	192.168.43.31
20	Licensing & Permits	192.168.43.32/27	255.255.255.224	192.168.43.33 - 192.168.43.62	192.168.43.33	192.168.43.63
30	Municipal Administration	192.168.43.64/27	255.255.255.224	192.168.43.65 - 192.168.43.94	192.168.43.65	192.168.43.95
40	Finance	192.168.43.96/27	255.255.255.224	192.168.43.97 - 192.168.43.126	192.168.43.97	192.168.43.127
50	Shared Print Services	192.168.43.128/28	255.255.255.240	192.168.43.129 - 192.168.43.142	192.168.43.129	192.168.43.143

Table 3: VLSM subnet summary for the five departmental / service networks.

Proposed router-to-router (WAN) addressing

The router brief's earlier serial addressing (192.168.43.72/30 and 192.168.43.76/30) falls inside the Municipal Administration subnet (192.168.43.64/27) under the revised design and cannot be used. The following unused address space is proposed instead and is treated as a proposed allocation pending confirmation in Packet Tracer:

Link	Network	Subnet mask	Router addresses	Broadcast
R1-R2 (proposed)	192.168.43.144/30	255.255.255.252	MUN-R1: .145 / MUN-R2: .146	192.168.43.147
R2-R3 (proposed)	192.168.43.148/30	255.255.255.252	MUN-R2: .149 / MUN-R3: .150	192.168.43.151

Table 4: Proposed point-to-point addressing for the router backbone.

4.4 End-Device Addressing Table

The table below sets out the proposed IP address for every end device in the topology, together with its department, VLAN and default gateway. Addresses are assigned sequentially after the reserved gateway address in each subnet. These addresses are proposed for Milestone 1 and should not be read as already configured in Packet Tracer unless confirmed by supporting evidence in the GitHub portfolio.

Device	Department	VLAN	IP address	Prefix	Default gateway
CS-PC01	Public Counter	10	192.168.43.2	/27	192.168.43.1
CS-PC02	Public Counter	10	192.168.43.3	/27	192.168.43.1
CS-PC03	Public Counter	10	192.168.43.4	/27	192.168.43.1
LP-PC01	Licensing & Permits	20	192.168.43.34	/27	192.168.43.33
LP-PC02	Licensing & Permits	20	192.168.43.35	/27	192.168.43.33
LP-PC03	Licensing & Permits	20	192.168.43.36	/27	192.168.43.33
ADM-PC01	Municipal Administration	30	192.168.43.66	/27	192.168.43.65
ADM-PC02	Municipal Administration	30	192.168.43.67	/27	192.168.43.65

Device	Department	VLAN	IP address	Prefix	Default gateway
ADM-PC03	Municipal Administration	30	192.168.43.68	/27	192.168.43.65
FIN-PC01	Finance	40	192.168.43.98	/27	192.168.43.97
FIN-PC02	Finance	40	192.168.43.99	/27	192.168.43.97
FIN-PC03	Finance	40	192.168.43.100	/27	192.168.43.97
PRINT-01	Shared Print Services	50	192.168.43.130	/28	192.168.43.129

Table 5: Proposed end-device IP addressing.

4.5 Router Interface Addressing

The table below summarizes the router-side interfaces required to support the VLAN gateways and the inter-router WAN links.

Router	Interface	Purpose	IP address	Prefix
MUN-R1	G0/0.10	VLAN 10 gateway	192.168.43.1	/27
MUN-R1	G0/0.20	VLAN 20 gateway	192.168.43.33	/27
MUN-R1	Serial link to MUN-R2	R1-R2 WAN	192.168.43.145	/30
MUN-R2	Serial link to MUN-R1	R1-R2 WAN	192.168.43.146	/30
MUN-R2	G0/0	VLAN 50 gateway	192.168.43.129	/28
MUN-R2	Serial link to MUN-R3	R2-R3 WAN	192.168.43.149	/30
MUN-R3	Serial link to MUN-R2	R2-R3 WAN	192.168.43.150	/30
MUN-R3	G0/0.30	VLAN 30 gateway	192.168.43.65	/27
MUN-R3	G0/0.40	VLAN 40 gateway	192.168.43.97	/27

Table 6: Proposed router interface addressing.

4.6 Addressing Justification

The addressing plan was designed to balance three considerations: efficient use of the assigned /24 block, sufficient headroom for departmental growth, and a clear, memorable numbering convention in which the first usable address in every subnet is reserved for the gateway. The /27 sizing for departments, the /28 sizing for the shared printer zone, and the /30 sizing for the WAN links each reflect the actual number of hosts required in that segment rather than a uniform subnet size, which keeps address wastage to a minimum across the 254 usable addresses in the assigned block.

5. Client Change Request

5.1 Shared Printer Requirement

Change request CR8 requires that a shared printer zone be established to serve two departments that currently cannot print.

5.2 Proposed Shared Printer Zone

Rather than placing PRINT-01 within a single department's VLAN, the design establishes a dedicated Shared Print Services network (VLAN 50) connected to MUN-R2. This centralized approach allows the printer to be accessed by authorized departments across the routed network while keeping it logically separate from individual departmental networks.

5.3 Departmental Access

The two departments selected to use the shared printer are Licensing & Permits (VLAN 20) and Municipal Administration (VLAN 30). This pairing demonstrates the purpose of the shared printer zone because the departments are located on different sides of the network, behind MUN-R1 and MUN-R3 respectively, while PRINT-01 is centrally connected through MUN-R2. Access to the printer will therefore also demonstrate successful communication across the multi-router network.

6. Design Justification

6.1 Device Selection

Three routers were selected to create the multi-router path required for the assigned **Static Routing** challenge. Each network area also uses its own access switch to connect local departmental devices and maintain a clear, organized network structure.

6.2 VLAN and Subnet Decisions

Each department is assigned a separate VLAN and /27 subnet to provide logical separation and support security controls between networks. The Shared Print Services network uses a smaller /28 subnet because it requires fewer host addresses, while /30 subnets are used efficiently for the point-to-point router links.

6.3 Routing Decisions

Static routing was selected because it directly satisfies the assigned technical challenge and is appropriate for the small, predictable three-router topology. It provides controlled communication between remote departmental networks without the additional complexity of a dynamic routing protocol.

6.4 Scalability and Future Growth

Each departmental /27 subnet provides **30 usable host addresses**, while currently supporting only three PCs and a gateway. This leaves sufficient capacity for additional workstations and other network devices. The dedicated Shared Print Services network can also accommodate additional shared-service devices in the future without redesigning the departmental subnets.

7. Security Design

7.1 Network Segmentation

The VLAN and subnet structure defined in Section 3.5 provides the logical foundation for controlling traffic between departments. VLAN segmentation creates separate broadcast domains but does not by itself prevent routed communication between them. An access control mechanism is therefore required to enforce the client's security constraint.

7.2 Public Counter Restriction

The client requires that Public Counter staff must not access the Municipal Administration network. The design therefore prevents traffic originating from Public Counter VLAN 10 (192.168.43.0/27) from accessing Administration VLAN 30 (192.168.43.64/27).

This restriction applies specifically to access from the Public Counter to Administration and does not require the Public Counter to be isolated from all other authorized networks.

7.3 Proposed ACL Strategy

An extended Access Control List (ACL) is proposed to enforce the restriction because it can filter traffic according to both source and destination networks. The ACL will deny traffic originating from VLAN 10 and destined for VLAN 30 while allowing other authorized communication.